

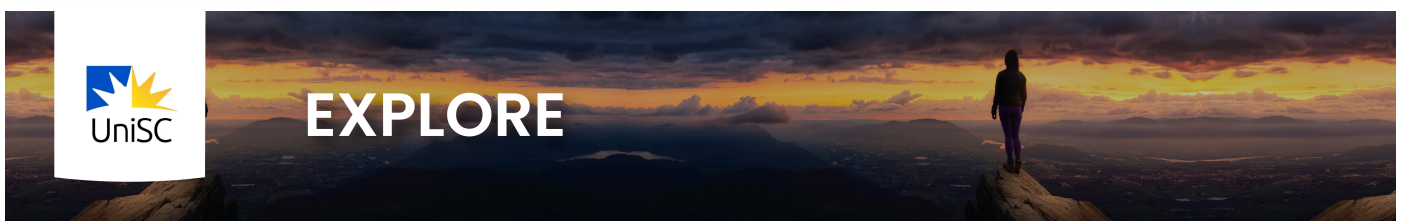
3.3 Risk-Adjusted Portfolio Performance $\updownarrow$



Risk-Adjusted Portfolio Performance

□ Guiding Questions

1. Why are returns alone insufficient for evaluating portfolio performance?
2. How can we determine whether a portfolio generated sufficient return relative to the risk taken?
3. What is the difference between the Sharpe Ratio and the Treynor Ratio?
4. When should total risk versus systematic risk be used when evaluating investment performance?



What will you learn?

You will be introduced to risk-adjusted performance evaluation. You will learn:

- Why high returns do not necessarily indicate strong investment performance.
- How the Sharpe Ratio evaluates returns relative to total portfolio volatility.

- How the Treynor Ratio evaluates returns relative to market (systematic) risk.
 - How risk-adjusted metrics allow fair comparison across portfolios with different risk levels.
 - How investors interpret these metrics when making portfolio allocation decisions.
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1. Why Do We Evaluate Risk-Adjusted Performance?

In previous sections, we learned how to measure portfolio returns and portfolio risk. However, evaluating return and risk separately does not fully capture the effectiveness of an investment strategy.

For example, a portfolio that generates a high return may appear attractive at first glance. However, if that return was achieved by taking extremely high risk, the investment may not be desirable.

Investors therefore need measures that evaluate how much return is earned relative to the level of risk taken. These are known as risk-adjusted performance measures.

Risk-adjusted measures allow investors to:

- compare portfolios with different risk levels
- determine whether higher returns compensate for higher risk
- evaluate portfolio manager performance
- identify more efficient investment strategies

These measures are widely used by portfolio managers, institutional investors, and financial analysts.

2. Excess Return and the Risk-Free Rate

Before measuring risk-adjusted performance, we need to understand the concept of excess return.

Investors could always choose to invest in a risk-free asset, such as government treasury securities. These investments offer relatively safe returns with minimal default risk.

Because investors can earn a risk-free return without taking significant risk, risky investments should provide additional compensation above this risk-free return.

This additional return is known as the risk premium or excess return.

Excess Return Formula

$$\text{Excess Return} = R_p - R_f$$

Where:

R_p = portfolio return

R_f = risk-free return

Excess return therefore represents the additional reward investors receive for bearing risk.

3. The Sharpe Ratio

The Sharpe ratio is one of the most widely used measures of risk-adjusted performance.

It measures the amount of excess return generated per unit of total risk.

Sharpe Ratio Formula

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

Where:

R_p = portfolio return

R_f = risk-free return

σ_p = portfolio standard deviation

The Sharpe ratio therefore combines both return and volatility into a single measure.



Watch

[Sharpe and Treynor Ratios](#)

The Big Why of Risk-Adjusted Performance

Why a bigger return doesn't always mean a better portfolio

PORTFOLIO A

14%

the bigger number

PORTFOLIO B

11%

the smarter one?

Details
Transcript

⋮

0:00 Well, let's talk about these two portfolios.

0:03 Portfolio A and portfolio B. Portfolio A returned

0:08 14% last year whereas portfolio B only returned

0:13 11%. Now, if I stopped right there asked you

4. Example: Calculating the Sharpe Ratio

Suppose a portfolio has the following characteristics:

Portfolio return = **12%**

Risk-free rate = **3%**

Portfolio standard deviation = **10%**

Step 1: Calculate excess return

$$12\% - 3\% = 9\%$$

Step 2: Divide by portfolio risk

$$\text{Sharpe} = \frac{9}{10} = 0.90$$

Interpretation

The portfolio generates 0.90 units of excess return for each unit of risk taken.

In general:

Sharpe Ratio	Interpretation
< 0.5	Weak risk-adjusted performance
0.5 – 1.0	Moderate performance
1.0 – 2.0	Strong performance
> 2.0	Exceptional performance

Higher Sharpe ratios indicate more efficient portfolios, meaning investors are receiving more return per unit of risk.

5. Comparing Portfolios Using the Sharpe Ratio

Consider two portfolios:

Portfolio	Return	Standard Deviation
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A	14%	12%
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B	11%	6%
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Assume the risk-free rate is **3%**.

Portfolio A

Excess return: $14 - 3 = 11$

Sharp ratio: $11/12 = 0.92$

Portfolio B

Excess return: $11 - 3 = 8$

Sharp ratio: $8/6 = 1.33$

Interpretation

Although Portfolio A has the higher return, Portfolio B has the higher Sharpe ratio.

This means Portfolio B provides more return per unit of risk and is therefore considered the more efficient portfolio.

This example illustrates why evaluating returns alone can be misleading.

6. The Treynor Ratio

Another commonly used measure of risk-adjusted performance is the Treynor ratio.

Unlike the Sharpe ratio, which uses total portfolio risk (standard deviation), the Treynor ratio focuses only on systematic risk, which is measured using beta.

Treynor Ratio Formula

$$\text{Treynor Ratio} = \frac{R_p - R_f}{\beta_p}$$

Where:

R_p = portfolio return

R_f = risk-free return

β_p = portfolio beta

Beta measures the sensitivity of a portfolio's returns to movements in the overall market.

A beta greater than 1 indicates higher market risk, while a beta below 1 indicates lower market sensitivity.

7. Example: Calculating the Treynor Ratio

Suppose a portfolio has:

Portfolio return = **13%**

Risk-free rate = **3%**

Portfolio beta = **1.2**

Step 1: Calculate excess return

$$13 - 3 = 10$$

Step 2: Divide by beta

$$\text{Treynor} = 10 / 1.2 = 8.33$$

Interpretation

The portfolio generates 8.33 units of excess return per unit of systematic risk.

This measure is particularly useful when comparing well-diversified portfolios, where unsystematic risk has largely been eliminated.

8. Differences Between the Sharpe and Treynor Ratios

Although both measures evaluate risk-adjusted performance, they differ in how risk is measured.

Measure	Risk Measure	When Used
Sharpe Ratio	Standard deviation (total risk)	When portfolios are not fully diversified
Treynor Ratio	Beta (systematic risk)	When portfolios are well diversified

Thus, the Sharpe ratio is often preferred when evaluating individual portfolios, while the Treynor ratio is more appropriate for diversified portfolios within broader investment strategies.

9. Limitations of Risk-Adjusted Performance Measures

Although risk-adjusted measures are useful, they also have limitations.

For example:

- Standard deviation treats positive and negative volatility equally
- Risk measures may depend on historical data that may not predict future risk
- Extreme events (financial crises) may not be captured accurately

Therefore, investors often combine several performance measures when evaluating investment strategies.

10. Key Insights from Risk-Adjusted Performance

From this section we learn:

- Returns alone cannot fully evaluate investment performance
- Investors must consider how much risk was taken to generate those returns
- Excess return measures the additional reward above the risk-free rate
- The Sharpe ratio evaluates return relative to total portfolio risk
- The Treynor ratio evaluates return relative to systematic market risk
- Risk-adjusted measures allow meaningful comparison between different portfolios

Risk-adjusted performance analysis therefore plays a crucial role in portfolio management and investment evaluation.

Reflect

What is meant by excess return?

Why must investors evaluate returns relative to risk?

What does the Sharpe ratio measure?

How does the Treynor ratio differ from the Sharpe ratio?

Why might a portfolio with lower returns still be considered superior?

Solve

Question 1

A portfolio has the following characteristics:

Portfolio return = 15%

Risk-free rate = 4%

Standard deviation = 11%

Calculate the **Sharpe ratio**.

Question 2

A portfolio has:

Portfolio return = 12%

Risk-free rate = 3%

Beta = 1.4

Calculate the **Treynor ratio**.

Question 3

Two portfolios are observed:

Portfolio Return Standard Deviation

A 13% 9%

B 11% 5%

Assume the risk-free rate is **2%**.

Calculate the Sharpe ratio for both portfolios and determine which performs better on a risk-adjusted basis.

Answers available [here \(https://learn.usc.edu.au/courses/35359/files/3292610?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3292610?wrap=1).

Critically evaluate

Explain why the Sharpe ratio may provide a more complete measure of risk-adjusted performance than the Treynor ratio in some cases.

Suppose a portfolio manager reports a very high return but also an extremely high level of volatility.

Discuss whether investors should consider this portfolio attractive and explain how risk-adjusted performance measures would help evaluate the investment.